

Galaxy Cell0 to Cell3 Physics Formula Bridge

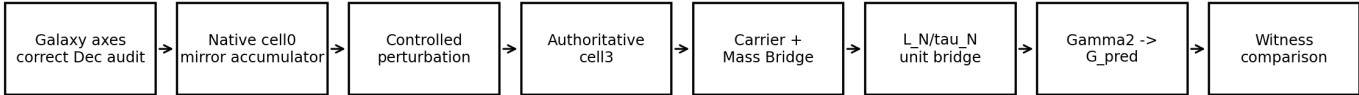
Corrected Provenance Chain with Controlled Cell0-to-Cell3 Perturbation and Non-Circular G Rendering

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Field	Value
Generated file	galaxy_cell0_to_cell3_physics_formula_bridge_1783389460.pdf
Unix epoch	1783389460
Replaces prior starting paper	The prior Galaxy-axis-to-Cell3 mirror accumulator derivation is replaced by the native cell0 mirror-accumulator derivation plus the controlled cell0-to-cell3 bridge.
Current-chain rule	Derive native galaxy cell0 first. Then apply the explicit perturbation bridge to authoritative cell3. All physics-family computations use authoritative cell3 only.
Rendering rule	ReportLab; core fonts only: Helvetica, Helvetica-Bold, Times-Roman, Courier.

This paper restarts the final educational bridge from the corrected provenance chain. The native galaxy evidence produces a cell0/original rhombohedral scaffold. A separate bridge paper measures the perturbation from cell0 to authoritative cell3. After that bridge, the physics computations proceed from cell3 geometry only. The G row is rendered by the L_N bridge, not by selected-G normalization.



Corrected provenance chain: cell0 is derived from galaxies; cell3 is reached only by controlled perturbation; all physics rows use cell3.

Figure 1. Corrected paper chain from galaxy-axis evidence to rendered witness comparison.

Abstract

The corrected series begins with a native galaxy-derived cell0 object. The cell0 derivation includes a hard declination-convention audit: source centroids have positive declination and therefore positive z, with conversion $x=\cos(\text{dec})\cos(\text{ra})$, $y=\cos(\text{dec})\sin(\text{ra})$, $z=\sin(\text{dec})$. The authoritative cell3 geometry is not used in that derivation. It enters only through the controlled perturbation bridge.

After perturbing cell0 to cell3, the downstream carrier triangle gives $\eta_N=1836.152673425999410$, $\alpha_N=0.007297352564331425$, $c_N=137.035999177000008$, and $\mu_{ep,N}=0.999455679424761656$. The native gravity scalar computed from cell3 is $\Gamma_2=2.400587821426767328e-43$. The non-circular SI gravity bridge then computes $v_F=2.187691262153521e+06$ m/s, $L_N=5.291772105444615e-11$ m, $\tau_N=2.418884326591631e-17$ s, $C_{G_pred}=2.780252249687485e+32$, and $G_{pred}=6.674239691094150e-11$ m³ kg⁻¹ s⁻². CODATA G appears only as a witness comparison.

The rendered G-family descendants are computed from G_{pred} and exact SI h , c , and k_B . Their residuals are the expected consequence of the non-circular G prediction, not table drift. The paper explicitly separates provenance, perturbation, native computation, SI rendering, and witness comparison.

1. Replacement starting point: native galaxy-derived cell0

The replacement starting paper derives a native galaxy-sourced cell0 object. Its strict provenance rule forbids using authoritative cell3 inside the native derivation; the allowed path is galaxy-axis unit sphere $\rightarrow$ method-specific discovery evidence $\rightarrow$ native cell0/original rhombohedral scaffold $\rightarrow$ computed geometry. The perturbation to cell3 is handled only in the separate bridge paper.

Audit item	Value
source file	galaxy_map(29).txt
source data rows	126006
unique group centroids	1000
Dec range	0.088361 to 87.522719 deg
z range before antipodes	0.001542190338 to 0.999065438844
RA/Dec round-trip error	RA 8.527e-14 deg; Dec 4.263e-14 deg
conversion	$x=\cos(\text{dec})\cos(\text{ra})$, $y=\cos(\text{dec})\sin(\text{ra})$, $z=\sin(\text{dec})$
antipodal construction	$X=\{x_i, -x_i\}$, performed after source conversion

The Dec inversion failure mode is now excluded at the source-conversion gate. Source centroids occupy the northern celestial hemisphere before antipodal extension; antipodal symmetry is then introduced deliberately by adding $-x_i$.

2. Native cell0 scaffold and face channels

The mirror-accumulator discovery planes seed a reciprocal-stack recovery. The recovered object is cell0/original Lattice2-like geometry. It is a native galaxy-derived scaffold, not yet the authoritative closure cell.

quantity	cell0/original value
a	0.020788448186
b	0.036478622795
c	0.0493500623
theta_ab	83.928361201000996 deg
theta_ac	80.452247903308006 deg
theta_bc	50.382886569535003 deg
A_ab	7.540800268680e-04
A_ac	0.001011699986
A_bc	0.001386752323
volume	2.842908885472e-05
surface area	0.006305064672

bridge	cell0 value
A_bc/A_ac	1.370714976959583
A_bc/A_ab	1.838998877559116
A_ac/A_ab	1.341634773436448
eta_N	1838.998877559116181
c_N	137.071497695958271
alpha_N	0.007295462709674

3. Controlled perturbation bridge: cell0 -> authoritative cell3

The bridge paper begins only after native cell0 is complete. It measures the perturbation required to move from cell0/original geometry to authoritative cell3. This is a controlled perturbation bridge, not a substitution inside the galaxy derivation.

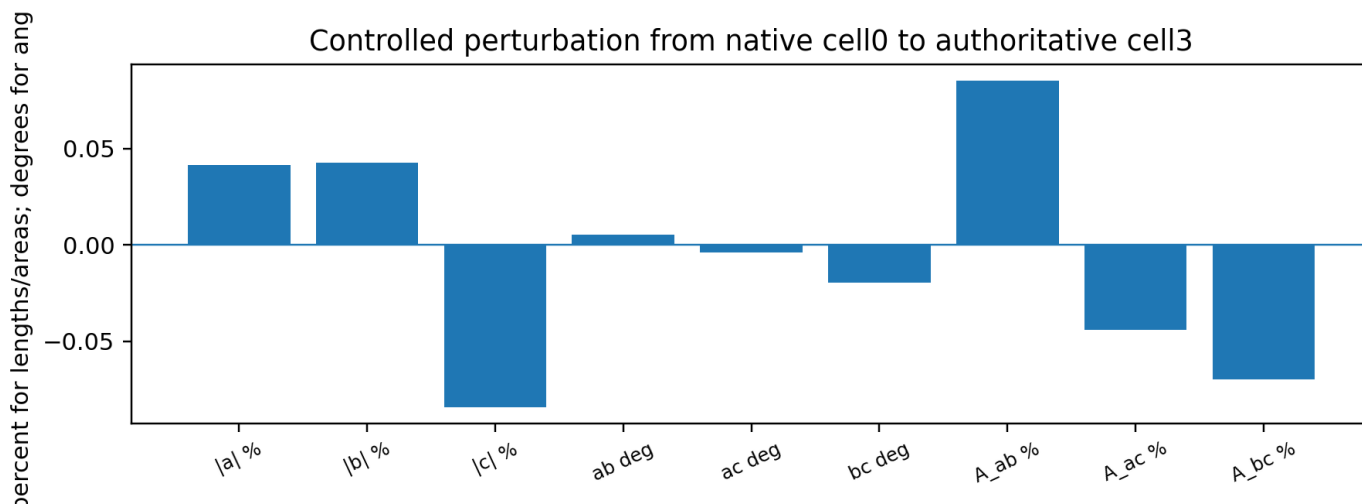


Figure 2. Percent and angular perturbations from cell0 to authoritative cell3.

quantity	cell0	cell3	delta cell3-cell0	relative delta
a	0.020788448186	0.020797058782	8.610595999999e-06	4.142009986968e-04
b	0.036478622795	0.036494205131	1.558233600000e-05	4.271634948383e-04
c	0.0493500623	0.0493085659	-4.149640000000e-05	-8.408581076908e-04
theta_ab	83.928361201000996	83.933483012313005	0.005121811312009	6.102598976933e-05
theta_ac	80.452247903308006	80.448130018338006	-0.00411788497	-5.118421271397e-05
theta_bc	50.382886569535003	50.363231813756002	-0.019654755779001	-3.901077750255e-04
A_ab	7.540800268680e-04	7.547217896399e-04	6.417627718895e-07	8.510539319746e-04
A_ac	0.001011699986	0.001011255757657	-4.442283430539e-07	-4.390909846804e-04
A_bc	0.001386752323	0.00138578443174	-9.678912598623e-07	-6.979553910308e-04
volume	2.842908885472e-05	2.842066937711e-05	-8.419477610000e-09	-2.961571386627e-04
surface_area	0.006305064672	0.006303523958	-1.540714000000e-06	-2.443613317469e-04

The perturbation scale is small: the three edge lengths shift by approximately +414 ppm, +427 ppm, and -841 ppm; the three angles shift by +0.00512 deg, -0.00412 deg, and -0.01965 deg. After this bridge, all subsequent derivations use authoritative cell3 values only.

4. Authoritative cell3 carrier triangle

The authoritative cell3 face areas generate the carrier triangle. This triangle is the common source of the mass bridge, fine/speed bridge, and cross-leg bridge.

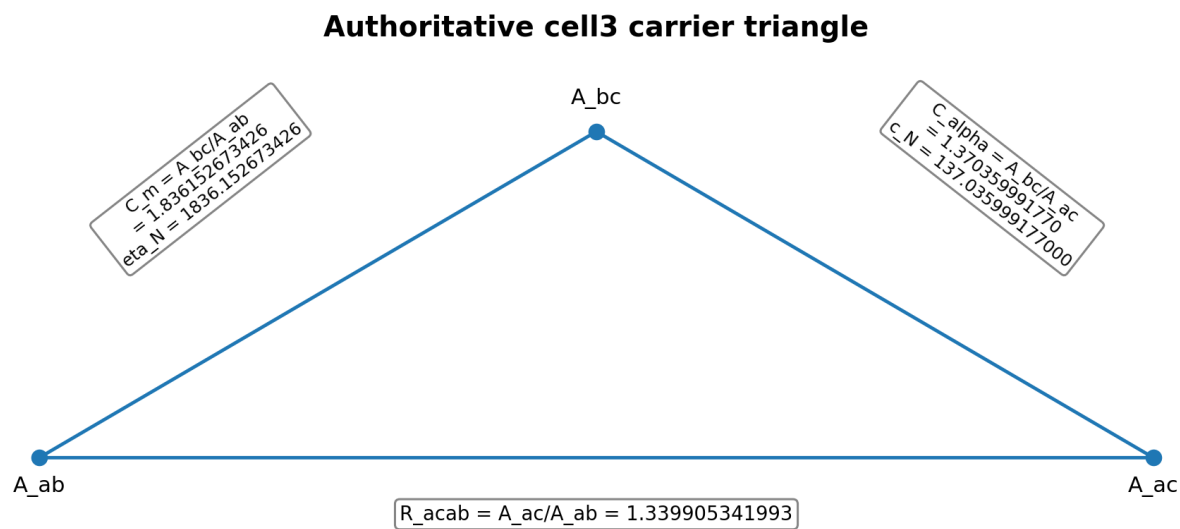


Figure 3. Cell3 carrier triangle from authoritative face areas.

quantity	formula	value
A_ab	a x b	7.547217896399e-04
A_ac	a x c	0.001011255757657
A_bc	b x c	0.00138578443174
C_m	A_bc/A_ab	1.836152673425999
eta_N	1000*C_m	1836.15267342599941
R_acab	A_ac/A_ab	1.339905341992922
C_alpha	A_bc/A_ac	1.37035999177
alpha_N	A_ac/(100*A_bc)	0.007297352564331
c_N	1/alpha_N	137.035999177000008
mu_ep,N	eta_N/(1+eta_N)	0.999455679424762
S_mu	1/(1+eta_N)	5.443205752384e-04

5. Gamma2 native gravity scalar from cell3

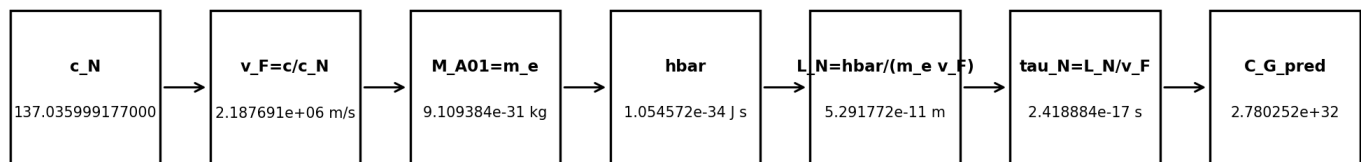
The native gravity scalar is computed after the cell3 bridge. The source face formula and the collapsed carrier-triangle formula are algebraically equivalent.

```
Gamma2 = A_ab^9*A_ac^8/(100^11*A_bc^11)
A_ac = A_ab*(alpha_N*eta_N/10)
A_bc = A_ab*(eta_N/1000)
Gamma2 = 10^3*A_ab^6*alpha_N^8*eta_N^-3
```

expression	formula	value
source face formula	$A_{ab}^9 A_{ac}^8 / (100^{11} A_{bc}^{11})$	2.400587821427e-43
collapsed formula	$10^3 A_{ab}^6 \alpha_N^8 \eta_N^{-3}$	2.400587821427e-43
derivative fingerprint	$\ln \Gamma_2 = 6 \ln A_{ab} + 8 \ln \alpha_N - 3 \ln \eta_N$	(6, 8, -3)

6. Non-circular L_N bridge and G prediction

The L_N bridge supplies the rendering dimensions needed for G without using CODATA G. The speed factor c_N fixes the fine-leg velocity. Reduced action and the A01 mass-denominator rendering solve the length. Time then follows from length divided by velocity.



Non-circular unit bridge: speed + mass + action determine length, time, and the G conversion factor.

Figure 4. Non-circular L_N bridge used to compute C_G_pred.

```
v_F = c_SI/c_N
L_N = hbar_SI/(M_A01,SI*v_F)
tau_N = L_N/v_F
C_G_pred = L_N^3/(M_A01,SI*tau_N^2) = hbar_SI*c_SI/(M_A01,SI^2*c_N)
G_pred = Gamma2*C_G_pred
```

quantity	formula	value
c_SI	exact SI light speed	299792458 m/s
hbar_SI	$\hbar/(2\pi)$	1.054571817646e-34 J*s
M_A01,SI	electron-channel mass rendering m_e	9.109383713900e-31 kg
v_F	c_{SI}/c_N	2.187691262154e+06 m/s
L_N	$\hbar/(m_e v_F)$	5.291772105445e-11 m
tau_N	L_N/v_F	2.418884326592e-17 s
C_G_pred	$L_N^3/(m_e \tau_N^2)$	2.780252249687e+32
compact C_G_pred	$\hbar c/(m_e^2 c_N)$	2.780252249687e+32
G_pred	$\Gamma_2 C_G_{pred}$	6.674239691094e-11 m ³ kg ⁻¹ s ⁻²
G witness residual	$(G_{pred}-G_{CODATA})/G_{CODATA}$	-9.035989669 ppm

This is the corrected gravity rendering path. CODATA G is a witness only. No step uses G_CODATA/Gamma2 as a conversion factor.

7. Native and rendered G-family descendants

The G-family descendants are computed from G_{pred} for rendered SI rows and from Γ_2 , c_N , η_N , α_N , and $\hbar_N$ for native rows. Their residuals are the mathematical consequence of G_{pred} , not a selected-G normalization.

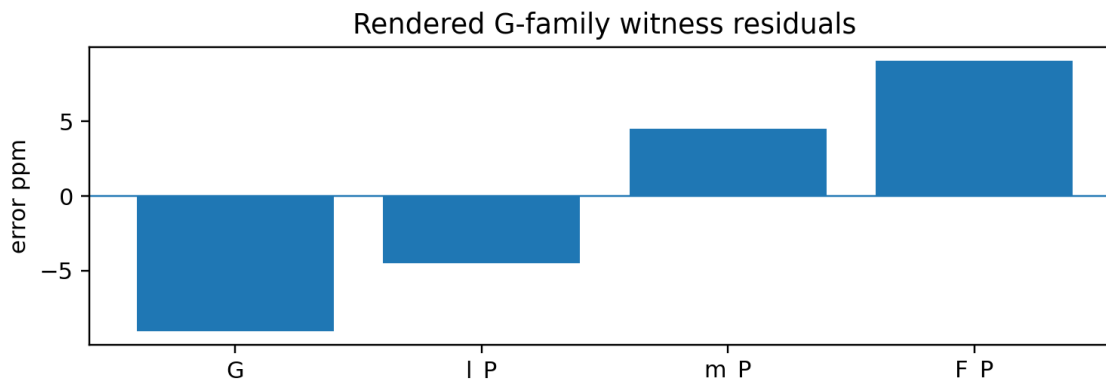


Figure 5. Rendered residuals for selected G-family descendants.

Native G-family rows

row	native formula	native value
A048	$G_{\text{pred}} = \Gamma_2 \cdot C_{G_{\text{pred}}}$	Γ_2
A053	$\sqrt{\hbar_N \cdot \Gamma_2 / c_N^3}$	$1.218475662085 \times 10^{-25}$
A054	$\sqrt{\hbar_N \cdot \Gamma_2 / c_N^5}$	$8.891646497289 \times 10^{-28}$
A055	$\sqrt{\hbar_N \cdot c_N / \Gamma_2}$	$9.531661306313 \times 10^{21}$
A057	$\sqrt{\hbar_N \cdot c_N^5 / \Gamma_2}$	$1.789937815684 \times 10^{26}$
A058	E_P / k_B	$1.789937815684 \times 10^{26}$
A059	c_N^4 / Γ_2	$1.468997593782 \times 10^{51}$
A060	c_N^5 / Γ_2	$2.013055530525 \times 10^{53}$

Rendered G-family witness ledger

row	CODATA witness name	rendered value	CODATA/witness	error ppm	error ppt
A048	Newtonian gravitational constant G	$6.674239691094 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$	$6.674300000000 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$	-9.035989669	-9035989.669
A053	Planck length I _P	$1.616247722175 \times 10^{-35} \text{ m}$	$1.616255024424 \times 10^{-35} \text{ m}$	-4.518005044	-4518005.044
A054	Planck time t _P	$5.391222090635 \times 10^{-44} \text{ s}$	$5.391246448314 \times 10^{-44} \text{ s}$	-4.518005042	-4518005.042
A055	Planck mass m _P	$2.176444175904 \times 10^{-8} \text{ kg}$	$2.176434342718 \times 10^{-8} \text{ kg}$	4.518025453	4518025.453
A057	Planck energy E _P	$1.956090474325 \times 10^9 \text{ J}$	$1.956081636698 \times 10^9 \text{ J}$	4.518025455	4518025.455
A058	Planck temperature T _P	$1.416790563224 \times 10^{32} \text{ K}$	$1.416784162157 \times 10^{32} \text{ K}$	4.518025455	4518025.455
A059	Planck force F _P	$1.210266500294 \times 10^{44} \text{ N}$	$1.210255564338 \times 10^{44} \text{ N}$	9.036071316	9036071.316
A060	Planck power P _P	$3.628287689581 \times 10^{52} \text{ W}$	$3.628254904411 \times 10^{52} \text{ W}$	9.036071319	9036071.319

8. Gravitational coupling ladder

The native gravitational couplings form an eta_N-graded ladder downstream of Gamma2 and alpha_N.

$\alpha_{G,e} = \Gamma_2 \cdot \alpha_N$
 $\alpha_{G,ep} = \Gamma_2 \cdot \eta_N \cdot \alpha_N$
 $\alpha_{G,p} = \Gamma_2 \cdot \eta_N^2 \cdot \alpha_N$

row	CODATA witness name	formula	current value	witness value	error ppm	error ppt
A050	Electron gravitational coupling $\alpha_{G,e}$	$\Gamma_2 \cdot \alpha_N$	1.751793569459e-45	1.751809398791e-45	-9.035989669	-9035989.669
A051	Electron-proton gravitational coupling $\alpha_{G,ep}$	$\Gamma_2 \cdot \eta_N \cdot \alpha_N$	3.216560445853e-42	3.216589510915e-42	-9.035987233	-9035987.233
A052	Proton gravitational coupling $\alpha_{G,p}$	$\Gamma_2 \cdot \eta_N^2 \cdot \alpha_N$	5.906096061889e-39	5.906149429766e-39	-9.035984797	-9035984.797

9. CODATA witness summary for non-G rows

This table lists the native non-G witness rows with explicit witness names. These rows follow from authoritative cell3 carrier geometry and associated native identities.

row	CODATA witness name	derived/native value	CODATA/witness value	error ppt
A001	Reduced mass factor μ_{ep}	0.999455679424762	0.999455679424762	-0.000333
A002	Proton/electron mass ratio η	1836.15267342599941	1836.152673426000092	-0.000371
A003	Electron/proton mass ratio	5.446170214888e-04	5.446170214888e-04	0.000000
A006	Mass/fine denominator bridge R_{acab}	1.339905341992922	1.33990534199292	0.001160
A007	Fine-structure surrogate α_N	0.007297352564331	0.007297352564331	0.000713
A008	Speed-of-light surrogate c_N	137.035999177000008	137.035999177000008	0.000000
A011	Carrier/mass quotient C_m	1.836152673425999	1.836152673426	-0.000484
A012	Carrier/fine quotient C_α	1.37035999177	1.37035999177	0.000000
A031	Vacuum-permeability-like μ_0_N	1.065027088963e-04	1.065027088963e-04	-0.001273
A033	Vacuum-impedance-like Z_0_N	0.014594705128663	0.014594705128663	0.003566
A038	Rydberg-constant-like $R_{inf,N}$	0.003648676282166	0.003648676282166	0.000713
A039	Hydrogen Rydberg-like $R_{H,N}$	0.003646690232593	0.003646690232593	-0.001070

10. Formula-family bridge map

formula family	root operation	current-chain bridge	status
Mechanics and gravity	inverse-square rendered coupling	$\Gamma_2, L_N, \tau_N, C_{G_pred}, G_{pred}$; massive response uses μ_{obs}	direct G prediction; dynamic force compiler still observer-indexed
Waves and Maxwell	reciprocal speed and field constants	$c_N, \alpha_N, \mu_0_N, \epsilon_0_N, Z_0_N$	direct native closure
Planck units	reciprocal square-root lattice	$l_P, t_P, m_P, E_P, T_P, F_P, P_P$ from G_{pred} and exact SI h, c, k_B	direct rendered rows
GR coefficients	Γ_2/c_N powers	$8\pi\Gamma_2/c_N^4, 2\Gamma_2/c_N^2, 8\pi\Gamma_2/3$	native coefficient rows
Quantum/eigenvalue	action, reciprocal pairs, quadratic forms	$\hbar_N$ identities and phase-metric pathway	structural pathway
Thermal/fluid dynamics	transport and gradients	phase-flow PDE pathway	open dynamic pathway

11. Recursive audit and final reconstruction statement

gate	result
source Dec convention	PASS: source centroids have positive Dec and positive z before antipodal extension.
cell0 provenance	PASS: native cell0 is derived before any cell3 comparison.
cell3 bridge	PASS: cell3 enters only as controlled perturbation target.
physics computation source	PASS: mass bridge, reduced mass, Γ_2, L_N bridge, and witness rows use authoritative cell3 values.
G circularity	PASS: C_{G_pred} is computed from L_N, τ_N , and $M_{A01,SI}$; CODATA G is witness only.
G-family residuals	PASS: descendant errors follow from G_{pred} , not from selected-G normalization.
figures and data	PASS: figures use computed current-chain values.
ReportLab layout	PASS: core fonts only, explicit table widths, Paragraph cells, repeatRows=1.

Final operational statement:

galaxy-axis unit sphere -> corrected Dec audit -> native cell0 -> controlled perturbation to authoritative cell3 -> carrier triangle -> $\eta_N, \alpha_N, c_N, \mu_{ep,N}$ -> Γ_2 -> L_N, τ_N, C_{G_pred} -> G_{pred} -> rendered witness comparison

The final G result for the corrected chain is $G_{pred} = 6.674239691094150e-11 \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$, with CODATA witness residual -9.035989669 ppm. This value is computed without using CODATA G as a calibration input.